Chapter 10. Nonparametric Inference

Wilcoxon rank-sum and signed-rank tests

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10.1 Introduction

Motivation. Classical t-based inference assumes normal data. When the data contain outliers or come from a heavy-tailed distribution, those procedures can lose much of their power, and methods built on signs and ranks are a safer alternative.

Most classical procedures assume the population is normal. If $X_1, \ldots, X_n$ is a random sample from $N(\mu, \sigma^2)$, then

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t(n-1),$$

and this pivot gives tests and confidence intervals for μ. Real populations are rarely exactly normal. The results still hold approximately because, by the central limit theorem, $\sqrt{n}(\overline{X} - \mu)/\sigma \to N(0,1)$ when n is large.

When the data contain outliers or the population is far from normal, normal-theory methods can fail badly. The actual confidence level can fall below the nominal one, the power of a test can drop sharply, and confidence intervals can become needlessly long.

Definition (Parametric, nonparametric and distribution-free methods)

(a) A **parametric method** assumes a specific parametric family (e.g. normal) for the population and makes inference about its parameters (μ, σ^2).
(b) A **nonparametric method** does not assume a parametric form for the distribution function. Its typical assumptions are only that the population distribution is *continuous*, or *continuous and symmetric*.
(c) A method that is valid whatever the population distribution is called **distribution-free**. For example, a test is distribution-free if the null distribution of its test statistic does not depend on the population distribution.

The two basic tools of nonparametric methods are the **sign** and the **rank**. The sign of an observation is 1 if it is positive and 0 if it is negative. The rank of an observation is its position when the observations are sorted in increasing order, counted from the smallest.

For example, the five observations

$$X_1 = 15, \quad X_2 = 18, \quad X_3 = 11, \quad X_4 = 36, \quad X_5 = 14$$

have ranks

$$R_1 = 3, \quad R_2 = 4, \quad R_3 = 1, \quad R_4 = 5, \quad R_5 = 2.$$

The largest value 36 gets rank 5 however large it is, so working with $R_1, \ldots, R_n$ instead of $X_1, \ldots, X_n$ limits the influence of outliers.

10.2 Wilcoxon rank-sum test

Motivation. Comparing two populations (or two treatments) is the *two-sample problem*. The Wilcoxon rank-sum test compares the locations of two distributions using only the ranks of the pooled sample.

Let the distribution functions of the two populations be $F(x)$ and $G(x) = F(x - \Delta)$. The two distributions have the same shape, and the Y distribution is the X distribution shifted by Δ. We call Δ the **shift parameter**. It can also be read as the difference of the means, $\Delta = \mu_Y - \mu_X$.

Definition (Data structure of the two-sample location problem)

$X_1, X_2, \ldots, X_m$: a random sample from a continuous distribution $F(x)$.
$Y_1, Y_2, \ldots, Y_n$: a random sample from the continuous distribution $F(x - \Delta)$.
The two samples are independent.

$\Delta = 0$ means that X and Y have the same distribution. $\Delta > 0$ means that the center of Y lies to the right of the center of X, and $\Delta < 0$ and $\Delta \neq 0$ are read in the same way.

Definition (Hypotheses on the shift parameter)

(i) $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$.
(ii) $H_0 : \Delta = 0$ versus $H_1 : \Delta < 0$.
(iii) $H_0 : \Delta = 0$ versus $H_1 : \Delta \neq 0$.

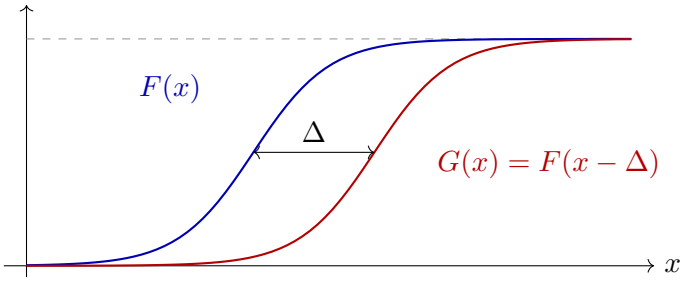


Figure 10.1: The distribution functions F and G under $H_1 : \Delta > 0$.

Pool the m observations X_i and the n observations Y_j into one sample of size $N = m + n$, the **combined sample**. Under $H_1 : \Delta > 0$ the Y's tend to be stochastically larger than the X's, so the Y observations tend to get larger ranks. A large sum of the Y-ranks in the combined sample is therefore evidence for H_1.

Definition 10-1 (Wilcoxon rank-sum statistic)

Let R_j be the rank of Y_j in the combined sample of the X's and Y's. The **Wilcoxon rank-sum statistic** is

$$W = \sum_{j=1}^{n} R_j.$$

10.2.1 Null distribution of W

To test with W we need its distribution under H_0. Take $m = 3$ and $n = 2$. Under $H_0 : \Delta = 0$ the X's and Y's have the same distribution, so $\{X_1, X_2, X_3, Y_1, Y_2\}$ is a random sample of size 5 from F. Assume there are no ties. Then all $\binom{5}{2} = 10$ arrangements of three x's and two y's in increasing order are equally likely.

No.	1	2	3	4	5	6	7	8	9	10
Arrangement	*yyxxx*	*yxyxx*	*yxxyx*	*yxxxy*	*xyyxx*	*xyxyx*	*xyxxy*	*xxyyx*	*xxyxy*	*xxxyy*
W	3	4	5	6	5	6	7	7	8	9

w	3	4	5	6	7	8	9	Total
$P(W = w)$	0.1	0.1	0.2	0.2	0.2	0.1	0.1	1.0

Tables 10-1 and 10-2: the ten arrangements (each with probability 0.1) and the null distribution of W for $m = 3$, $n = 2$.

The null distribution of W does not involve F, so the rank-sum test is distribution-free. For instance, with $m = 3$, $n = 2$ and $\alpha = 0.10$, the rejection region for $H_1 : \Delta > 0$ is $W \geq 9$, whatever F is.

The null distribution of W can be computed for every m and n. Tables of it (Appendix C of the book) or software give the tail probabilities. In the table the smaller sample is taken as the Y sample, and W_s denotes its rank sum. The table lists right-tail probabilities $P(W_s \geq x)$ and left-tail probabilities $P(W_s \leq x^*)$. For example, with smaller sample size $n = 3$ and larger sample size $m = 7$, $x = 25$ and $x^* = 8$ give

$$P(W_s \geq 25) = 0.033 = P(W_s \leq 8).$$

(The null distribution is symmetric about $n(N+1)/2 = 16.5$, so $x^* = n(N+1) - x$.)

Definition (Wilcoxon rank-sum test)

Take the larger sample as the X sample and the smaller one as the Y sample, and let

$$W = \text{sum of the ranks of the } Y \text{ sample in the combined sample.}$$

The rejection regions for $H_0 : \Delta = 0$ at level α are as follows.
(i) For $H_1 : \Delta > 0$: reject if $W \geq c$.
(ii) For $H_1 : \Delta < 0$: reject if $W \leq c^*$.
(iii) For $H_1 : \Delta \neq 0$: reject if $W \geq c_1$ or $W \leq c_1^*$.
The critical values satisfy

$$P(W \geq c) = \alpha, \qquad P(W \leq c^*) = \alpha, \qquad P(W \geq c_1) = \frac{\alpha}{2} = P(W \leq c_1^*).$$

Example 10-1 (Fine dust in two cities)

Cities A and B have similar population density and geography. Their monthly mean PM_{10} concentrations ($\mu g/m^3$) from March to November were recorded.

Month	3	4	5	6	7	8	9	10	11
City A	33	28	23	25	18	17	25.5	27	26
City B	22	18.5	19.5	20	16	24	22.5	21.5	21

At the 5% level, use the rank-sum test to decide whether the two cities differ.
Solution. Test $H_0 : \Delta = 0$ against $H_1 : \Delta \neq 0$, with $X = A$ and $Y = B$ ($m = n = 9$). The ranks in the combined sample of 18 values are listed below.

X_i	33	28	23	25	18	17	25.5	27	26
rank	18	17	11	13	3	2	14	16	15
Y_j	22	18.5	19.5	20	16	24	22.5	21.5	21
rank	9	4	5	6	1	12	10	8	7

Hence $W_s = 9 + 4 + 5 + 6 + 1 + 12 + 10 + 8 + 7 = 62$. From the table with $m = n = 9$,

$$P(W_s \geq 109) = 0.02 = P(W_s \leq 62).$$

The two-sided p-value is $2 \times 0.02 = 0.04 < 0.05$. We reject H_0 and conclude that the PM_{10} levels of the two cities differ.

Example 10-2 (Tire wear)

Tires of company A cost about 20% more than those of company B. Eight tires of each company were driven 10,000 km and their wear rates (%) were measured. One A measurement was lost.

Company A	15	22	27	18	13	19	25	
Company B	20	16	17	26	28	24	30	33

Can we say that tires of A are better than those of B?
Solution. The sample from A is smaller, so A is the Y sample ($n = 7$) and B is the X sample ($m = 8$). Better quality means lower wear, so we test $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$.

X_i (B)	16	17	20	24	26	28	30	33
rank	3	4	7	9	11	13	14	15
Y_j (A)	13	15	18	19	22	25	27	
rank	1	2	5	6	8	10	12	

So $W_s = 44$, and the table with $m = 8$, $n = 7$ gives $P(W_s \le 44) = 0.095$. The p-value 9.5% is above 5% and below 10%. We cannot reject H_0 at the 5% level, but at the 10% level we conclude that tires of A are better.

10.2.2 Ties and the normal approximation

So far there were no ties. Even for a continuous population, real data are rounded, so ties occur. Tied observations get the average of the ranks they occupy (**midranks**), and W_s is computed with these. The tabulated null distribution then holds only approximately, so the test becomes an approximate test.

When m and n are too large for the tables, we use a normal approximation. We standardize W by its null mean and variance and refer the result to $N(0,1)$, which is a Z test.

Theorem 10-1 (Normal approximation of W)

Under $H_0 : \Delta = 0$,

$$E(W) = \frac{n(m+n+1)}{2}, \qquad \mathrm{Var}(W) = \frac{mn(m+n+1)}{12}.$$

When m and n are large,

$$Z_W = \frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Write $N = m + n$ and assume there are no ties.

Step 1. Under H_0 the N observations are i.i.d. from a continuous distribution. Every ordering of them is therefore equally likely, and the vector $(R_1, \dots, R_n)$ of Y-ranks is a random ordered selection of n values drawn without replacement from $\{1, \dots, N\}$. In particular each R_j is uniform on $\{1, \dots, N\}$, and each pair (R_j, R_k) with $j \neq k$ is uniform on the ordered pairs of distinct values.

Step 2. For a uniform R on $\{1, \dots, N\}$,

$$E(R) = \frac{1}{N}\sum_{r=1}^{N} r = \frac{N+1}{2},$$

$$\mathrm{Var}(R) = \frac{1}{N}\sum_{r=1}^{N} r^2 - \Big(\frac{N+1}{2}\Big)^2 = \frac{(N+1)(2N+1)}{6} - \frac{(N+1)^2}{4} = \frac{N^2-1}{12}.$$

Step 3. By exchangeability $c = \mathrm{Cov}(R_j, R_k)$ is the same for all $j \neq k$. If all N observations are ranked, the ranks sum to the constant $N(N+1)/2$, whose variance is 0:

$$0 = N\,\frac{N^2-1}{12} + N(N-1)\,c, \qquad \text{so} \qquad c = -\frac{N+1}{12}.$$

Step 4. Hence $E(W) = n(N+1)/2 = n(m+n+1)/2$ and

$$\begin{aligned}\mathrm{Var}(W) &= n\,\frac{N^2-1}{12} - n(n-1)\,\frac{N+1}{12} = \frac{n(N+1)}{12}\big[(N-1)-(n-1)\big] \\ &= \frac{n(N+1)(N-n)}{12} = \frac{mn(m+n+1)}{12}.\end{aligned}$$

Step 5. W is the total of a simple random sample of size n drawn without replacement from the finite population $\{1,\dots,N\}$. By the central limit theorem for sampling without replacement (Wald–Wolfowitz, Hájek), such a total is asymptotically normal when $\min(n,m) \to \infty$, because no single population value dominates the population variance. Therefore $Z_W \xrightarrow{d} N(0,1)$. ■

Example 10-3 (Normal approximation for Example 10-2)

Use the normal approximation of W for the data of Example 10-2.
Solution. There $W = 44$ with $m = 8$ and $n = 7$, so

$$E(W) = \frac{7(8+7+1)}{2} = 56, \qquad \mathrm{Var}(W) = \frac{8\times 7\times(8+7+1)}{12} = 74.67,$$
$$Z_W = \frac{44-56}{\sqrt{74.67}} = -1.39.$$

$P(Z \le -1.39) = 0.0823$, so the one-sided p-value is 8.23%. This is very close to the exact result of Example 10-2.

Example 10-4 (New surgical method)

A new surgical method A is compared with the standard method B. Each method was applied to 10 patients, and the recovery times (days) are given below.

A	8	9	11	12	16	17	19	19	21	45
B	13	16	18	18	20	20	21	23	25	72

To decide whether A is more effective than B, carry out
(a) the t test;
(b) the Wilcoxon rank-sum test;
(c) the normal-approximation rank-sum test.
Solution. Shorter recovery means a more effective method. With B as the Y sample, we test $H_0 : \Delta = 0$ against $H_1 : \Delta > 0$.
(a) For A, $\bar{x} = 17.7$ and $\sum(x_i - \bar{x})^2 = 1010.1$. For B, $\bar{y} = 24.6$ and $\sum(y_i - \bar{y})^2 = 2600.4$. Then

$$s_p^2 = \frac{1010.1 + 2600.4}{18} = 200.6, \qquad s_p = 14.16,$$
$$t = \frac{\bar{y} - \bar{x}}{s_p\sqrt{1/m + 1/n}} = \frac{24.6 - 17.7}{14.16\sqrt{1/10 + 1/10}} = 1.09.$$

Since $t_{0.1}(18) = 1.330 > 1.09$, the p-value exceeds 10%, and the t test cannot conclude that A is more effective.
(b) With midranks for the ties, the ranks in the combined sample are as follows.

X_i (A)	8	9	11	12	16	17	19	19	21	45
rank	1	2	3	4	6.5	8	11.5	11.5	15.5	19
Y_j (B)	13	16	18	18	20	20	21	23	25	72
rank	5	6.5	9.5	9.5	13.5	13.5	15.5	17	18	20

So $W_s = 128$. With $m = n = 10$ the table gives $P(W_s \geq 128) = 0.045$. At the 5% level, A is more effective than B.
(c) We have

$$E(W) = \frac{10 \times 21}{2} = 105, \qquad \mathrm{Var}(W) = \frac{10 \times 10 \times 21}{12} = 175,$$
$$Z_W = \frac{128 - 105}{\sqrt{175}} = 1.74,$$

and $P(Z \geq 1.74) = 0.0409$. The p-value is 4.09%, so again A is more effective than B.

Why the t test failed here

In Example 10-4 the rank-sum test gives $p \approx 4$–4.5%, but the t test gives $p > 10\%$. The cause is the two outliers 45 and 72, which inflate s_p and weaken the t test. If these were recording errors or came from exceptional patients and were removed, then

$$\bar{x} = 14.67, \quad \sum(x_i - \bar{x})^2 = 182.00, \qquad \bar{y} = 19.33, \quad \sum(y_i - \bar{y})^2 = 104.00,$$
$$s_p^2 = \frac{182.00 + 104.00}{16} = 17.88, \qquad s_p = 4.23, \qquad t = \frac{19.33 - 14.67}{4.23\sqrt{1/9 + 1/9}} = 4.957.$$

Since $t_{0.0005}(16) = 4.015$, the p-value is below 0.05%, and A is clearly better. With outliers or markedly non-normal data, the rank-sum test is often more efficient than the t test.

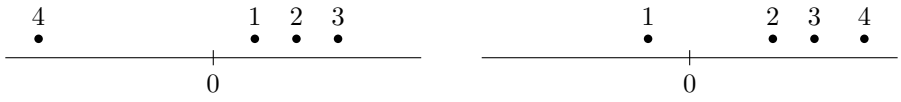
10.3 Wilcoxon signed-rank test

Motivation. For the center of a single population, the t test of Section 6.1 targets the mean. The signed-rank test targets the center of a symmetric distribution using only the signs and the ranks of the absolute values, and it also handles paired comparisons.

Definition (Data structure of the one-sample location problem)

$X_1, X_2, \ldots, X_n$ is a random sample from a distribution $F(x - \theta)$ that is continuous and symmetric about θ. The **location parameter** θ is the center of the distribution, which is both its mean (if it exists) and its median.

Consider two samples of $n = 4$ observations (Figure 10.2). Both have three positive observations. In (a) the negative observation lies far from 0, while in (b) it lies close to 0.



(a) negative value relatively far from 0 (b) negative value relatively close to 0

Figure 10.2: Two samples with $n = 4$; the numbers are the absolute ranks.

The **absolute rank** of an observation is its rank by distance from 0: regardless of sign, values closer to 0 get smaller ranks. To test whether the location is 0, we can use the sum of absolute ranks of the positive observations:

$$\text{(a) } T = 1 + 2 + 3 = 6, \qquad \text{(b) } T = 2 + 3 + 4 = 9.$$

The farther the data are shifted to the right of 0, the larger the ranks or the number of the positive observations, so this sum grows. In general, to test $H_0 : \theta = \theta_0$ we set

$$Z_i = X_i - \theta_0 \qquad (i = 1, \dots, n).$$

Under H_0, $Z_1, \dots, Z_n$ follow a distribution symmetric about 0. Let R_i^+ be the absolute rank of $|Z_i|$ among $|Z_1|, \dots, |Z_n|$. The **Wilcoxon signed-rank statistic** is

$$W^+ = \text{sum of the absolute ranks } R_i^+ \text{ of the positive } Z_i.$$

Under H_0 the Z_i are symmetric about 0, so positive and negative Z_i have the same distribution of absolute ranks (uniform), and each sign is $+$ or $-$ with probability $\frac{1}{2}$. Hence the null distribution of W^+ does not depend on F, and the signed-rank test is distribution-free.

The book's Appendix C tabulates both tails of W^+: right-tail probabilities $P(W^+ \geq x)$ and left-tail probabilities $P(W^+ \leq x^*)$. For $n = 4$, for instance, $P(W^+ \geq 9) = 0.125 = P(W^+ \leq 1)$. Indeed, among the $2^4 = 16$ equally likely sign patterns, only $\{2, 3, 4\}$ and $\{1, 2, 3, 4\}$ have rank sum at least 9.

Definition (Wilcoxon signed-rank test)

(i) Compute $Z_i = X_i - \theta_0$ for $i = 1, \dots, n$.
(ii) Find the absolute rank R_i^+ of $|Z_i|$ among $|Z_1|, \dots, |Z_n|$.
(iii) Compute $W^+ =$ the sum of R_i^+ over the positive Z_i.
The rejection regions for $H_0 : \theta = \theta_0$ are as follows.
For $H_1 : \theta > \theta_0$: reject if $W^+ \geq c$.
For $H_1 : \theta < \theta_0$: reject if $W^+ \leq c^*$.
For $H_1 : \theta \neq \theta_0$: reject if $W^+ \geq c_1$ or $W^+ \leq c_1^*$.
The critical values satisfy

$$P(W^+ \geq c) = \alpha, \quad P(W^+ \leq c^*) = \alpha, \quad P(W^+ \geq c_1) = \frac{\alpha}{2} = P(W^+ \leq c_1^*).$$

Example 10-5 (Smartwatch heart-rate error)

A smartwatch maker claims that the average error of its heart-rate measurement is 5 bpm, where the error is the absolute difference between the true and the measured rates. A consumer group suspects the error exceeds 5 bpm. It tested 10 randomly chosen watches on 10 people of similar build against a medical monitor, obtaining the following errors (bpm).

3.5	2.6	7.8	6.0	6.8	3.1	8.8	8.5	7.7	4.5

Is the consumer group's claim supported?

Solution. Test $H_0 : \theta = 5$ against $H_1 : \theta > 5$. With $\theta_0 = 5$, the values $Z_i = X_i - 5$ are

$$-1.5,\ -2.4,\ 2.8,\ 1.0,\ 1.8,\ -1.9,\ 3.8,\ 3.5,\ 2.7,\ -0.5.$$

negative Z_i	abs. rank	positive Z_i	abs. rank
-0.5	1	1.0	2
-1.5	3	1.8	4
-1.9	5	2.7	7
-2.4	6	2.8	8
		3.5	9
		3.8	10

So $W^+ = 2+4+7+8+9+10 = 40$. With $n = 10$ the table gives $P(W^+ \geq 40) = 0.116$. The p-value 11.6% is above 5%, so the evidence is insufficient to conclude that the error exceeds 5 bpm.

When values of the observations or of $|Z_i|$ are tied, we use average ranks. As with the rank-sum statistic, the null distribution of W^+ then holds only approximately. When n is too large for the table, we use the normal approximation of W^+.

Theorem 10-2 (Normal approximation of W^+)

Under $H_0 : \theta = \theta_0$,

$$E(W^+) = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(W^+) = \frac{n(n+1)(2n+1)}{24}.$$

When n is large,

$$Z_{W^+} = \frac{W^+ - E(W^+)}{\sqrt{\mathrm{Var}(W^+)}}$$

approximately follows the standard normal distribution $N(0,1)$.

Proof. [Supplement] Assume there are no ties, and let $Z = X - \theta_0$ have a continuous distribution symmetric about 0 under H_0.

Step 1. The sign of Z is independent of $|Z|$. For $a > 0$, symmetry gives

$$P(Z > 0,\ |Z| \leq a) = P(0 < Z \leq a) = \tfrac{1}{2} P(|Z| \leq a) = P(Z > 0)\, P(|Z| \leq a).$$

Step 2. For $k = 1, \ldots, n$ let $I_k = 1$ if the observation with absolute rank k is positive, and $I_k = 0$ otherwise. The absolute ranks depend only on $|Z_1|, \ldots, |Z_n|$, and by Step 1 the signs are i.i.d. fair coin flips independent of these. Hence $I_1, \ldots, I_n$ are i.i.d. Bernoulli($\frac{1}{2}$), and

$$W^+ = \sum_{k=1}^{n} k\, I_k.$$

Step 3. With $E(I_k) = \frac{1}{2}$ and $\mathrm{Var}(I_k) = \frac{1}{4}$,

$$E(W^+) = \frac{1}{2} \sum_{k=1}^{n} k = \frac{n(n+1)}{4},$$

$$\mathrm{Var}(W^+) = \frac{1}{4} \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{24}.$$

Step 4. The summands $Y_k = k(I_k - \tfrac{1}{2})$ are independent with mean 0 and $|Y_k| = k/2$. Write $s_n^2 = \mathrm{Var}(W^+)$. Then

$$\frac{\sum_{k=1}^{n} E|Y_k|^3}{s_n^3} = \frac{\tfrac{1}{8}\sum k^3}{s_n^3} = \frac{n^2(n+1)^2/32}{\big(n(n+1)(2n+1)/24\big)^{3/2}} = O(n^{-1/2}) \to 0.$$

This is Lyapunov's condition, so the Lyapunov CLT gives $Z_{W^+} \xrightarrow{d} N(0,1)$. ■

10.3.1 Paired comparison

Section 6.2 treated paired comparison for the difference of two means. If the experimental units are very heterogeneous, it is better to group them into homogeneous pairs than to split them at random into two groups. With n pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ we form the differences

$$D_i = Y_i - X_i \qquad (i = 1, \dots, n).$$

The hypotheses $H_0 : \Delta = 0$ versus $H_1 : \Delta > 0$ on the shift parameter become $H_0 : \theta = 0$ versus $H_1 : \theta > 0$ on the location of the D_i. Applying the signed-rank test to the D_i gives a nonparametric paired comparison.

Example 10-6 (Twins and a new curriculum)

To make statistics easier to learn, a new curriculum A was developed. Researchers recruited 16 pairs of identical twins with equal test scores. One twin of each pair took the old curriculum B and the other took A, and both then sat the same test.

A	70	85	73	75	65	80	71	80
B	65	43	45	80	84	71	52	42
$Z = A - B$	5	42	28	−5	−19	9	19	38
A	51	72	76	79	59	84	56	57
B	78	62	38	80	54	87	38	43
$Z = A - B$	−27	10	38	−1	5	−3	18	14

At the 5% level, is the new curriculum more effective?

Solution. Let θ be the location of Z, and test $H_0 : \theta = 0$ against $H_1 : \theta > 0$. Absolute ranks (with midranks for ties) are as follows.

negative Z_i	−1	−3	−5	−19	−27
abs. rank	1	2	4	10.5	12

positive Z_i	5	5	9	10	14	18	19	28	38	38	42
abs. rank	4	4	6	7	8	9	10.5	13	14.5	14.5	16

So $W^+ = 106.5$. With $n = 16$, Theorem 10-2 gives

$$E(W^+) = \frac{16 \times 17}{4} = 68, \qquad \mathrm{Var}(W^+) = \frac{16 \times 17 \times 33}{24} = 374,$$
$$Z_{W^+} = \frac{106.5 - 68}{\sqrt{374}} = 1.99.$$

$P(Z \ge 1.99) = 0.0233$, so the one-sided p-value 2.33% is below 5%. We reject H_0 and conclude that the new curriculum A is more effective than B.